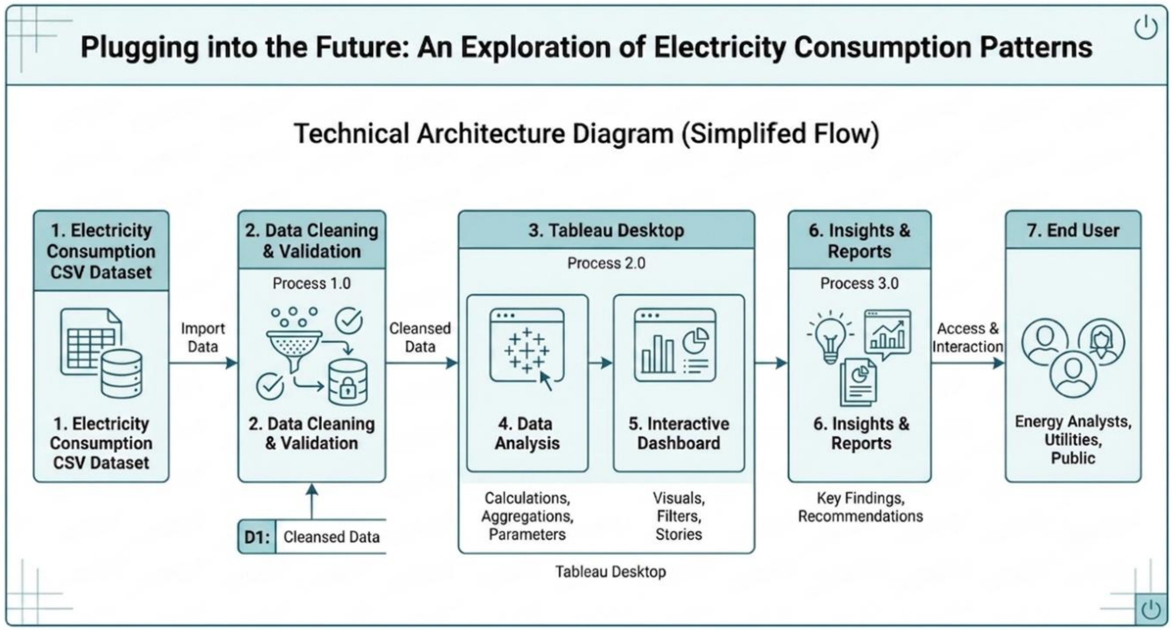
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2



2.	Application Logic-1	Data Cleaning and Data Analysis	**Tableau Desktop**
3.	Application Logic-2	Data Visualization and Interactive Dashboard Creation	**Tableau Desktop**
4.	Application Logic-3	Dashboard Filtering and Report Generation	**Tableau Desktop**
5.	Database	Electricity Consumption Dataset (CSV File)	CSV Dataset (Local File)
6.	Cloud Database	Not used	N/A
7.	File Storage	Store the dataset and Tableau workbook	Local File System (.csv, .twb/.twbx)
8.	External API-1	Not used	N/A
9.	External API-2	Not used	N/A
10.	Machine Learning Model	Not used	N/A
11.	Infrastructure (Server / Cloud)	Dashboard published online for viewing	Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Data visualization and dashboard publishing platform used for analyzing electricity consumption data.	Tableau Public
2.	Security Implementations	Access to the project is controlled through Tableau Public account authentication. The dataset is stored locally before publishing.	Tableau Public Authentication
3.	Scalable Architecture	The architecture supports analyzing larger datasets and additional years by importing updated CSV files into Tableau without changing the dashboard design.	Tableau Desktop + CSV Dataset
4.	Availability	The published dashboard is available online through Tableau Public and can be accessed from any device with an internet connection	Tableau Public
5.	Performance	The dashboard processes the electricity consumption dataset efficiently and generates interactive visualizations with minimal loading time. Filters and charts respond quickly for smooth data analysis.	Tableau Desktop, Tableau Public